Atefeh Rohani

Applied Scientist / Formal Verification & ML Reliability for Safety-Critical Systems / PhD in Computer Science

Formal Verification Engineer & AI Research Scientist with focus on ML Reliability for Safety-Critical Systems. With more than 6 years work experience, I am expert in developing predictive AI models, specializing in time-series data analysis, deep learning, and MLOps pipelines.

Education

Aug 2025 – Now **(Certificate in Advanced Studies) CAS in Advanced Data Science and Machine Learning**, *Universität Bern*, Switzerland, Topics: Scientific Computing and Machine Learning, Deep Learning

Sep 2019 – Aug 2023 **PhD in Computer Science**, *Universität Bern*, Switzerland,
Thesis title: Conditional Obligations in Justification Logic

Sep 2012 – Aug 2014 **M.Sc. in Soft Computing**, *Shahid Bahonar University of Kerman*,
Thesis title: Non-axiomatic Logic, a Model of Intelligent Reasoning

Sep 2006 – Aug 2010 **B.Sc. in Mathematics**, *Shahid Bahonar University of Kerman*,
Thesis title: An Introduction to Neural Networks

Sep 2002 – Aug 2006 **High school Diploma**, *Mathematics and Physics*, Kerman,
Farzanegan High school, administered by National Organization of Exceptional Talents

Work Experiences

Research Experiences

Apr 2025 – Now **Senior Researcher**, *Software Engineering Group*, Institut für Informatik, Universität Bern, Switzerland

Focus: AI/ML Engineering for Safety-Critical Systems, MLOps, and Predictive Maintenance.

Current project (Funded by Haslerstiftung): Advancing Unmanned Aerial Vehicle Reliability and Societal Trust Through Integrated Testing and Formal Verification:

- **Predictive AI Development:** Architected and deployed end-to-end predictive models using Python,TensorFlow, and Scikit-learn to forecast system failures in UAVs, utilizing large-scale time-series flight logs to significantly reduce critical error rates.

Hilfikerstrasse 2 – 3014 Bern – Switzerland
+41 76 749 78 24 • *atefeh.rohani@unibe.ch* • *atefeh-rohani*

- **Anomaly Detection & Specification Mining:** Engineered unsupervised learning pipelines to automatically identify misbehaviors and anomalous patterns in autonomous systems, enhancing the resilience and safety of UAV flight operations.
- **ML Reliability & Testing:** Integrated formal verification with machine learning testing frameworks to ensure the high-assurance and reliability of AI-driven safety-critical software architectures.
- **Project Leadership in R&D:** Spearheaded multidisciplinary research projects at the intersection of Data Science, Robotics, and Formal Methods, managing the transition from simulated model training to real-world drone flight experiments.

Mar 2024 – Mar 2025 **Post Doctoral Researcher**, *Software Engineering Group*, Institut für Informatik, Universität Bern, Switzerland

Finished project: Runtime Verification small-scale flight and space software This project investigates runtime verification techniques for space software to increase system's assurance.

- Implemented Formal Verification and Runtime Verification techniques to ensure high-assurance and reliability in safety-critical software architectures (e.g., Space Software Architectures).
- Developed formal specification languages (logical systems) to precisely model and verify software requirements for high-assurance, safety-critical systems (e.g., satellites).
- **Requirement Formalization:** Developed and evaluated temporal logic frameworks to transform complex, real-world software requirements into precise mathematical specifications for time-sensitive ML systems.

Sep 2019 – Aug 2023 **Doctoral Researcher**, *Logic and Theory Group*, Institut für Informatik, Universität Bern, Switzerland

Fields of Research: Mathematical Logic, Temporal logic, Proof Theory, Automated Theorem Proving, Logic and Computation, Computability and Complexity Theory.

Projects:

Engineered logical frameworks to formally represent operational and ethical *norms* for Explainable AI (XAI) systems, enhancing model interpretability and compliance.

- Utilized a deep theoretical understanding of Advanced Algorithms and Complexity Theory to optimize the performance and scalability of AI solutions.
- Providing mathematical proofs for soundness and completeness of our novel logical systems
- Thesis topic: (Conditional Obligations in Justification Logic) Using the novel system for representing conditional obligations, which are a new representation of conditionals in normative systems.

Additional activities:

- Member of Swiss Graduate Society for Logic and Philosophy of Science, Organizing workshop in University of Bern, October 2022
- Webmaster of the website of logic and theory group, Institute of Computer Science, University of Bern

Teaching Experiences

Sep 2020 – Feb 2023 **Teaching Assistant**, *Universität Bern*, Bern, Switzerland

Lectures:

- **Justification Logic**, for Master students of mathematics and computer science
- **Advanced Algorithms**, for Master students of computer science
- **Computability and Complexity**, for Bachelor students of mathematics and computer science

Sep 2014 – Aug 2019 **Teaching mathematics at high school**, *Kerman*, Teaching mathematics high schools in my hometown.

Sep 2010 – Aug 2012 **Teaching experiences and voluntary activities**, *Kerman*, Voluntary activities in an NGO called "Mathematics' Home" with the aim of publicising science, mathematics especially.

Languages

English	Fluent
German	C1
Persian	Mother tongue
Arabic	Basic

Skills

Bar Skills	**Quantitative Research, Formal Verification, Algorithm Development Software Verification & Testing Methods, ML Reliability & Robustness, Formal Methods in Software Engineering, Probabilistic reasoning, Logical Formalization, Software Requirements Engineering, Algorithm Development, Data Engineering, Data Mining, Data Analysis, Machine Learning, Deep Learning, Mathematical Models in Machine Learning, Academic and Scientific Writing**
IT Skills	**Python, Libraries: Pandas, NumPy, Libraries: Scikit-learn, TensorFlow, PyTorch, JAX, Libraries: Matplotlib, Seaborn, SciPy, Keras, Azure, MATLAB, Simulink, Stateflow, GitHub, R-Studio, HTML, CSS, LaTeX**
Management Skills	**Research Management, Interdisciplinary Research, Organization Skills, Presentation Skills**
Formal Verification:	**SMT Solvers: Z3 / CVC5, Model Checkers: TLA+ / UPPAAL, Verification Languages: Coq / Lean / Isabelle**
Testing & Quality:	**Runtime Verification, Specification Mining, System Resilience,**
Safety-Critical AI:	**Predictive Maintenance, Verified AI/ML, Time-series Analysis, Signal Processing, Requirement Engineering,**
Data Science:	**Object-oriented Programming, AI-based Learning, Machine Learning, Deep Learning, Gen-AI: LLMs / Azure, Mathematical Models in Data Science**
Soft Skills	**Project Management, Interpersonal Communication, Technical Translation, Detail Orientation, Teamwork, Stakeholder Communication, Communication in International Environment, Critical Thinking and Analytical Skills, Creative Problem-Solving, Perseverance and Resilience, Intellectual Curiosity, Scientific Rigor, Adaptability & Continuous Learning**
presentation Skills:	**Structured and Logical Content Delivery, Confident Use of Multimedia Tools**

Publications

- Explicit Non-normal Modal Logics, joint work with Thomas Studer, Journal of Logic and Computation 2021.
- Explicit Non-normal Modal Logics, joint work with Thomas Studer, International Workshop on Logic, Language, Information, and Computation, WoLLIC 2021.

- Deontic Conditional in Justification Logic, Joint work with Federico Faroldi and Thomas Studer, Proceeding of international Workshop on Logic, Language, Information and Computation, WoLLIC 2023.
- Temporal Logics Meet Real-World Software Requirements: A Reality Check, Joint work with Roman Bögli, Thomas Studer, Christos Tsigkanos, and Timo Kehrer, In Proceedings of the 2025 IEEE/ACM 13th International Conference on Formal Methods in Software Engineering (FormaliSE), Ottawa, Canada, 2025.

Awards

March 2024 **Research proposal awarded by the Haslerstiftung**, The purpose of the Hasler Foundation is to promote information and communications technology (ICT) for the well-being and benefit of Switzerland as an intellectual and industrial center. The Foundation aims to play an active role in ensuring that Switzerland continues to take a leading position in science and technology in the future.

Membership

Sep 2020 – Aug 2023 **Member of Swiss Graduate Society for Logic and Philosophy of Science**

Activities:

- Organising workshop in university of Bern, October 2022
- Editor of the website for Swiss Society for logic and Philosophy of Science, under SCNAT (Swiss Academy of Science) domain

Interests

Singing Member of Bahar Choir (Soprano), Bahar Choir consists of Iranian singers, performing with the East-Paris Philarmonic Orchestra

Hiking & Biking Swiss Alps

References

- **Dr. Sebastiano Panichella**
  Software Engineering Group, Institute of Computer Science, University of Bern
  email: sebastiano.panichella@unibe.ch
  Website: https://spanichella.github.io
- **Prof. Dr. Thomas Studer**
  Logic and Theory Group, Institute of Computer Science, University of Bern
  email: thomas.studer@unibe.ch
  Website: https://home.inf.unibe.ch/ tstuder/
  Phone: Direct: +41 31 684 39 84 Secretary: +41 31 684 84 26
- **Prof. Dr. Timo Kehrer**
  Software Engineering Group, Institute of Computer Science, University of Bern
  email: timo.kehrer@unibe.ch
- **Prof. Dr. Federico Faroldi**
  Department of Law, University of Pavia, Italy
  email: federico.faroldi@unipv.it